

Event Overview

The Mission:Brain Global Neurosurgery Hackathon is an innovation competition challenging multidisciplinary teams to develop solutions that improve neurosurgical and TBI care for patients in low- and middle-income countries (LMICs). You have been given a detailed clinical case vignette describing a patient's full journey - from unrecognized chronic brain injury through acute neurosurgical emergency to long-term disability - set in rural Pakistan. Your task is to identify a specific problem within this narrative and design a feasible, impactful solution.

Before You Begin

Read the case vignette carefully and in full. The case is intentionally comprehensive. It spans prevention, prehospital care, acute diagnosis, emergency surgery, postoperative management, rehabilitation, and systemic barriers. You are not expected to address the entire story. The strongest submission will focus on one (or at most two) specific pain points and develop a deep, well-reasoned, and evidence-backed solution.

A list of suggested pain points is provided at the end of the case vignette. You may choose from this list or identify your own problem within the narrative. If you identify your own, be prepared to justify why it represents a meaningful opportunity for intervention.

What to Build

Your deliverable is a business pitch of your solution concept. This may take the form of a device, a protocol, a training program, a digital tool, a policy framework, a community intervention, or any other approach that addresses your chosen pain point. Your solution should be:

- Specific: clearly define the problem you are solving and for whom
- Feasible: designed for the real-world constraints described in the case - limited electricity, no internet reliability, minimal trained personnel, extreme poverty, cultural and geographic barriers
- Evidence-informed: grounded in what is known about TBI care, global surgery, or LMIC health systems. You are encouraged to cite literature, guidelines, or existing models that support your approach
- Scalable or adaptable: consider whether your solution could be applied beyond this single case to other rural settings, other LMICs, or other neurological emergencies

Judges are looking for an evidence-backed concept with a clear implementation and validation plan.

What Not to Do

- Do not try to solve the entire case. A narrow, well-developed solution is far stronger than a broad, superficial one
- Do not assume unlimited resources. Solutions requiring reliable high-speed internet, continuous electricity, specialist physicians, or expensive equipment must justify how these dependencies would be met in the described setting

- Do not ignore the human context. The case describes a real pattern of injury, care, and disability experienced by millions of people. Solutions should reflect awareness of cultural, economic, and social realities, not just technical feasibility

Resources

You are free to use any publicly available resources during the competition, including medical literature, open-source tools, and online databases. Mentors will be available throughout the event for clinical questions, technical guidance, and feedback on your approach.

Our team has prepared and collected several resources to help you begin tackling the problem, which are available within the Participant Access Drive folder.

<https://drive.google.com/drive/folders/1ogmlfnrlbqaBlgku9AzCHxFKf3CNWYHk?usp=sharing>

A Note on Scope

Traumatic brain injury kills or disables millions of people each year, with the vast majority of the burden falling on LMICs where access to neurosurgical care is severely limited. The case you will read is fictional, but every element - the unrecognized chronic injury, the delayed presentation, the difficult transport, the improvised surgery, the absent rehabilitation - reflects documented and real-world realities. Your work this weekend has the potential to generate ideas that matter beyond this room. Go make someone's life better.